IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

ELECTROMAGNETISM

Wednesday, 8 May 2:30 pm

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : O. Sydoruk
Second Marker(s) : R. Syms

© Imperial College London

Formula sheet

Universal constants: $\varepsilon_0 = 8.84 \times 10^{-12}$ F/m $\quad \mu_0 = 4\pi \times 10^{-7}$ A/m $\quad c = 3 \times 10^8$ m/s

Maxwell's equations in integral form

$$\oiint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s} = \iiint_V \rho \mathrm{d}v$$
$$\oiint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s} = 0$$
$$\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{l} = \iint_S \mathbf{J} \cdot \mathrm{d}\mathbf{s} + \frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s}$$
$$\oint_L \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s}$$

Maxwell's equations in differential form

$$\nabla \cdot \mathbf{D} = \rho$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Constitutive relations: $\quad \mathbf{D} = \varepsilon_0 \varepsilon_\mathrm{d} \mathbf{E} \qquad \mathbf{B} = \mu_0 \mu_\mathrm{r} \mathbf{H} \qquad \mathbf{J} = \sigma \mathbf{E}$

Boundary conditions

$$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \sigma \qquad (\mathbf{B}_1 - \mathbf{B}_2) \cdot \mathbf{n} = 0$$
$$\mathbf{n} \times [\mathbf{E}_1 - \mathbf{E}_2] = 0 \qquad \mathbf{n} \times [\mathbf{H}_1 - \mathbf{H}_2] = \mathbf{J}_\mathrm{s}$$

Solution of Poisson's equation for electrostatic potential

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \iiint_V \frac{\rho(\mathbf{r}')\mathrm{d}v}{|\mathbf{r} - \mathbf{r}'|}$$

Biot-Savart law

$$\mathbf{B} = \frac{\mu_0}{4\pi} I \int_L \frac{\mathrm{d}\mathbf{l} \times \mathbf{r}}{r^3}$$

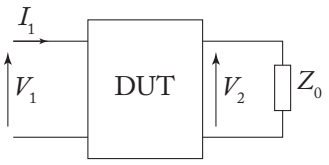
Transmission-line equations

$$\frac{\partial I(z,t)}{\partial z} = -C \frac{\partial V(z,t)}{\partial t} \qquad \text{and} \qquad \frac{\partial V(z,t)}{\partial z} = -L \frac{\partial I(z,t)}{\partial t}$$

Input impedance of a loaded transmission line

$$Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan(kd)}{Z_0 + jZ_{\text{L}} \tan(kd)}$$

S-parameters for a two-port device



$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \quad \text{and} \quad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1}$$

Fields of a short dipole antenna (far-field region)

$$E_\theta = jZ_0 \frac{I_0 kd}{4\pi} \frac{e^{-jkr}}{r} \sin\theta$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Fields of a half-wave dipole antenna (far-field region)

$$E_\theta = \frac{jZ_0 I_0}{2\pi} \frac{\cos\dfrac{\pi\cos\theta}{2}}{\sin\theta} \frac{e^{-jkr}}{r}$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Gauss' and Stokes' theorems

$$\oiint_S \mathbf{F} \cdot \mathrm{d}\mathbf{s} = \iiint_V \nabla \cdot \mathbf{F}\, \mathrm{d}v \qquad \oint_L \mathbf{F} \cdot \mathrm{d}\mathbf{l} = \iint_S [\nabla \times \mathbf{F}] \cdot \mathrm{d}\mathbf{s}$$

Useful integrals

$$\int \frac{\mathrm{d}x}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{a^2 + x^2}) \quad \int \frac{\mathrm{d}x}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2 + a^2}} \quad \int \frac{x\,\mathrm{d}x}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}}$$

ELECTROMAGNETISM

1. a) An infinitely long straight conductor carries a dc current. Sketch the magnetic field lines, assuming the current flows out of the plane of paper. Referring to Maxwell's equations, show that reversing the direction of the current reverses the direction of the magnetic field lines. [5]

b) Referring to Maxwell's equations, show how the concept of voltage is introduced for static electric fields. [10]

c) Referring to Maxwell's equations, outline difficulties arising in definition of voltage for time-varying electric fields. [10]

d) Figure 1.1 shows a two-port device. The ports have impedances Z_0. [10]

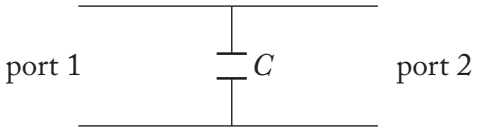


Figure 1.1

i) Find the S-parameters.

ii) Comment on power conservation.

iii) Sketch the variation of the magnitudes of the S-parameters with frequency, indicating any significant points.

iv) What function does the device perform?

e) Show that the power flow of a short dipole antenna in the far-field region is radial. [5]

2. a) An infinitely large plane is carrying a uniform dc current with a linear density J, as shown in Fig. 2.1(a). The plane is surrounded by air. Assume that the magnitude of the magnetic field strength created by this current does not depend on the distance away from the plane.

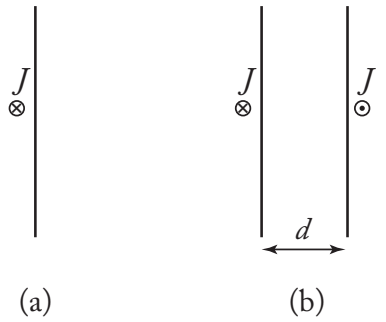


Figure 2.1

i) Sketch the magnetic field lines. Justify your answer. [5]

ii) Find the magnetic field strength. [10]

b) Two infinitely large planes are carrying uniform dc currents of equal linear density J but in the opposite directions, as shown in Fig. 2.1(b). The distance between the planes is d.

i) Find the magnetic field strength. [5]

ii) Find the inductance per unit length of two parallel plates, assuming each has a width b. [10]

3. Two perfectly conducting parallel plates are separated by an air gap of a width d, as shown in Fig. 3.1.

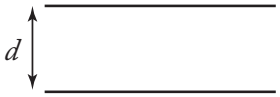


Figure 3.1

a) Assuming plates of a finite width b that have per-unit-length capacitance C and inductance L, sketch an equivalent circuit that shows that the plates can act as a transmission line. Assuming that $C = \varepsilon_0 b/d$ and the wave velocity is $c = 1/\sqrt{\varepsilon_0 \mu_0}$, find L. [10]

b) Briefly describe how wave propagation along the transmission line can be interpreted as propagation of a transverse electromagnetic wave. [10]

c) Sketch a dispersion diagram (i.e. a variation of ω with k) indicating the transverse electromagnetic (TEM), transverse magnetic (TM), and transverse electric (TE) waves. Briefly and without derivations, explain the reason TE and TM modes have frequency cut-off. [10]